Spirometer



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Spirometer

What it is used for

A spirometer measures lung volumes or airflow during breathing tests. It is used for supervised respiratory assessment or training, and results must be interpreted by a qualified healthcare professional.

Main parts

- Identify the spirometer body and display or measurement scale.
- Locate the mouthpiece and breathing tube or flow sensor.
- Identify the filter or disposable breathing accessory, where provided.
- Locate the start, stop and reset controls.
- Identify the calibration syringe connection or calibration function, if applicable.
- Locate the power supply or battery compartment.
- Identify the manufacturer's instructions and patient-use accessories.

Before you start

1. Use the spirometer under appropriate healthcare or trained instructor supervision.
2. Read the instructions and confirm the device is suitable for the intended test.
3. Inspect the mouthpiece, tubing, sensor and housing for damage or contamination.
4. Clean hands and use a new disposable mouthpiece or filter when specified.
5. Check battery level, device settings and calibration status according to instructions.
6. Explain the procedure and obtain the person's consent before testing.
7. Ensure the person is seated comfortably and understands the breathing instructions.

How to use

1. Switch on the spirometer and complete the required startup checks.
2. Attach a clean mouthpiece and filter according to the device instructions.
3. Select the prescribed test mode and enter required patient details accurately.
4. Position the person upright and fit a nose clip if the test protocol requires one.
5. Ask the person to seal their lips around the mouthpiece without obstructing airflow.
6. Demonstrate and instruct the breathing manoeuvre specified for the test.
7. Start recording and encourage the person to follow the clinician's breathing instructions.
8. Repeat the manoeuvre only as directed, allowing rest between attempts when needed.
9. Review the recorded values and check for errors or an invalid test indication.
10. Save the results securely and remove the mouthpiece after testing.
11. Clean reusable components and switch off the device according to instructions.

⚠ Safety

- Do not perform forced breathing tests without appropriate supervision and screening.
- Stop the test if the person develops chest pain, severe breathlessness, dizziness or distress.
- Never share mouthpieces without appropriate cleaning or a new disposable accessory.
- Use filters and infection-control procedures according to device instructions.
- Do not interpret spirometry values as a diagnosis without qualified clinical assessment.
- Follow healthcare guidance for people with relevant medical risks or recent procedures.

Care & storage

- Clean and disinfect reusable components according to manufacturer instructions.
- Replace disposable mouthpieces and filters after each user as required.
- Check calibration and perform quality-control procedures at the recommended intervals.
- Store the device and accessories in a clean, dry and protected case.

Common mistakes to avoid

- Using a dirty or incorrectly fitted mouthpiece or filter.
- Failing to explain the required breathing manoeuvre before recording.
- Accepting poor-quality or invalid test results without review.
- Changing calibration settings or interpreting results without appropriate training.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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